

Mathematics A

Unit 1 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

68 classified items • 259 marks • 19 chapter(s)

Source-faithful practice with synchronized solutions

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Unit 1 • Expertise • Unit 1 • Expertise Q16+ • 68 items • 259 marks

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June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Set Notation & Venn Diagrams

Unit 1 • Expertise • 7 marks • 3 item(s)

January 2021 | 4MA1-2H Q16 | 3 marks

Complete or solve a practical three-set survey

January 2021 | 4MA1-2H Q16 | 3 marks

Complete or solve a practical three-set survey

June 2021 | 4MA1-1H Q19 | 1 marks

Read and combine three-set regions

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Fractions, Decimals & Percentages

Unit 1 • Expertise • 7 marks • 3 item(s)

June 2025 | 4MA1-2H Q18 | 2 marks

Pure recurring block

November 2021 | 4MA1-1H Q18 | 3 marks

Non-recurring prefix followed by a recurring block

January 2023 | 2H Q16 | 2 marks

Decimal to fraction or percentage

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Rounding, Estimation & Bounds

Unit 1 • Expertise • 16 marks • 4 item(s)

June 2025 | 4MA1-2H Q22 | 4 marks

Geometry, rate or practical decision bounds

November 2024 | 4MA1-1H Q21 | 3 marks

Algebraic expression upper or lower bound

May-Nov 2020 | 1H Q17 | 4 marks

Mixed stated accuracies

January 2023 | 1H Q19 | 5 marks

Round to decimal places or significant figures

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Surds

Unit 1 • Expertise • 14 marks • 5 item(s)

June 2023 | 4MA1-2H Q23 | 4 marks

Binomial surd denominator using a conjugate

November 2024 | 4MA1-1H Q17 | 1 marks

Extract square factors from a surd

June 2022 | 4MA1-1H Q16 | 3 marks

Add or subtract like surds

January 2020 | 4MA1-1H Q17 | 3 marks

Expand binomials containing surds

May-Nov 2020 | 1H Q19 | 3 marks

Single-term surd denominator

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Algebraic Roots & Indices

Unit 1 • Expertise • 11 marks • 3 item(s)

June 2024 | 4MA1-2H Q23 | 3 marks

Complex indexed monomial or fractional outer power

June 2022 | 4MA1-2H Q24 | 3 marks

Express one exponent variable in terms of others

January 2023 | 2H Q26 | 5 marks

Zero and negative indices

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Expanding Brackets

Unit 1 • Expertise • 3 marks • 1 item(s)

November 2023 | 4MA1-2H Q16 | 3 marks

Three linear brackets

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Factorising

Unit 1 • Expertise • 2 marks • 1 item(s)

November 2021 | 4MA1-1H Q17 | 2 marks

Non-monic quadratic ax^2+bx+c

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Completing the Square

Unit 1 • Expertise • 7 marks • 2 item(s)

June 2023 | 4MA1-2H Q19 | 3 marks

Positive non-unit leading coefficient

January 2022 | 4MA1-1H Q20 | 4 marks

Negative leading coefficient or maximum form

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Algebraic Fractions

Unit 1 • Expertise • 18 marks • 4 item(s)

November 2023 | 4MA1-2H Q24 | 5 marks

Equation containing algebraic fractions

November 2024 | 4MA1-2H Q23 | 4 marks

Divide factorisable algebraic fractions

January 2021 | 4MA1-2H Q21 | 4 marks

Multiply factorisable algebraic fractions

May-Nov 2020 | 1H Q21 | 5 marks

Substitute a variable relation into a fraction

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Coordinate Geometry

Unit 1 • Expertise • 23 marks • 4 item(s)

June 2022 | 4MA1-2H Q22 | 7 marks

Kite or symmetric vertex coordinates

January 2020 | 4MA1-1H Q22 | 6 marks

Coordinates from isosceles and gradient constraints

January 2021 | 4MA1-2H Q18 | 4 marks

Rectangle or parallelogram missing vertex

January 2023 | 1H Q23 | 6 marks

Read a plotted coordinate pair

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Linear Graphs ($y = mx + c$)

Unit 1 • Expertise • 10 marks • 3 item(s)

June 2024 | 4MA1-2H Q18 | 2 marks

Parallel and perpendicular gradient relation

January 2020 | 4MA1-2H Q19 | 1 marks

Parallel line through a point

November 2021 | 4MA1-1H Q20 | 7 marks

Perpendicular line through a point or midpoint

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Graphs of Functions

Unit 1 • Expertise • 6 marks • 2 item(s)

June 2025 | 4MA1-2H Q25 | 3 marks

Feature-led polynomial sketch

June 2023 | 4MA1-2H Q18 | 3 marks

Recognise reciprocal or rational graph shape

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Area & Perimeter

Unit 1 • Expertise • 7 marks • 2 item(s)

January 2021 | 4MA1-2H Q18 | 3 marks

Find perimeter of a simple triangle or rectangle

January 2021 | 4MA1-2H Q20 | 4 marks

Find composite or shaded polygonal area

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Circles, Arcs & Sectors

Unit 1 • Expertise • 28 marks • 6 item(s)

November 2023 | 4MA1-2H Q20 | 4 marks

Find annular, overlapping or multi-circle area

June 2023 | 4MA1-1H Q22 | 6 marks

Solve an algebraic circle-measure constraint

November 2024 | 4MA1-1H Q16 | 4 marks

Find sector area

November 2024 | 4MA1-2H Q25 | 5 marks

Find annular, overlapping or multi-circle area

January 2022 | 4MA1-1H Q16 | 4 marks

Recover radius or central angle from an arc/sector measure

January 2023 | 1H Q21 | 5 marks

Find a circular segment area or perimeter

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Sine, Cosine Rule & Area of Triangles

Unit 1 • Expertise • 26 marks • 5 item(s)

June 2025 | 4MA1-1H Q20 | 5 marks

Use two non-right triangles sharing a diagonal

June 2023 | 4MA1-2H Q22 | 5 marks

Combine non-right trigonometry with circle or elevation geometry

June 2022 | 4MA1-1H Q18 | 5 marks

Find an acute angle using the sine rule

January 2022 | 4MA1-2H Q23 | 6 marks

Resolve the ambiguous sine-rule angle

January 2023 | 1H Q16 | 5 marks

Find a side using the sine rule

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

3D Pythagoras & Trigonometry

Unit 1 • Expertise • 19 marks • 4 item(s)

January 2022 | 4MA1-1H Q21 | 5 marks

Use midpoint or symmetry structure in a 3D model

June 2021 | 4MA1-1H Q22 | 4 marks

Use a non-rectangular base before a line-plane angle

May-Nov 2020 | 2H Q18 | 5 marks

Find angle between a line and a plane

January 2023 | 2H Q23 | 5 marks

Use 3D trigonometry to obtain a terminal volume

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Histograms

Unit 1 • Expertise • 8 marks • 4 item(s)

June 2022 | 4MA1-1H Q21 | 3 marks

Estimate a partial-class frequency

June 2021 | 4MA1-1H Q18 | 2 marks

Complete a table from a partial histogram

June 2021 | 4MA1-1H Q18 | 1 marks

Complete missing histogram bars from mixed information

June 2021 | 4MA1-1H Q18 | 2 marks

Estimate a histogram proportion, fraction or percentage

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Probability Diagrams - Venn & Tree Diagrams

Unit 1 • Expertise • 11 marks • 3 item(s)

June 2024 | 4MA1-2H Q16 | 7 marks

Read regions and probabilities from a Venn diagram

November 2021 | 4MA1-1H Q16 | 2 marks

Solve unknown Venn regions using a total

November 2021 | 4MA1-1H Q16 | 2 marks

Complete a three-set Venn diagram

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Combined & Conditional Probability

Unit 1 • Expertise • 36 marks • 9 item(s)

June 2025 | 4MA1-1H Q21 | 3 marks

Calculate an exact composition in three or more draws

June 2025 | 4MA1-2H Q20 | 6 marks

Calculate a direct given-that probability

June 2024 | 4MA1-1H Q16 | 4 marks

Use a complement in repeated no-replacement selection

November 2023 | 4MA1-1H Q20 | 4 marks

Find exactly one, same or different independent outcomes

June 2022 | 4MA1-2H Q16 | 8 marks

Calculate a direct given-that probability

January 2020 | 4MA1-2H Q18 | 2 marks

Use a histogram or table as a conditional sample space

November 2021 | 4MA1-1H Q16 | 1 marks

Calculate a direct given-that probability

November 2021 | 4MA1-2H Q16 | 3 marks

Compare with-replacement and without-replacement models

May-Nov 2020 | 2H Q20 | 5 marks

Recover an unknown population from no-replacement probability

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

16 Some students were asked the following question.

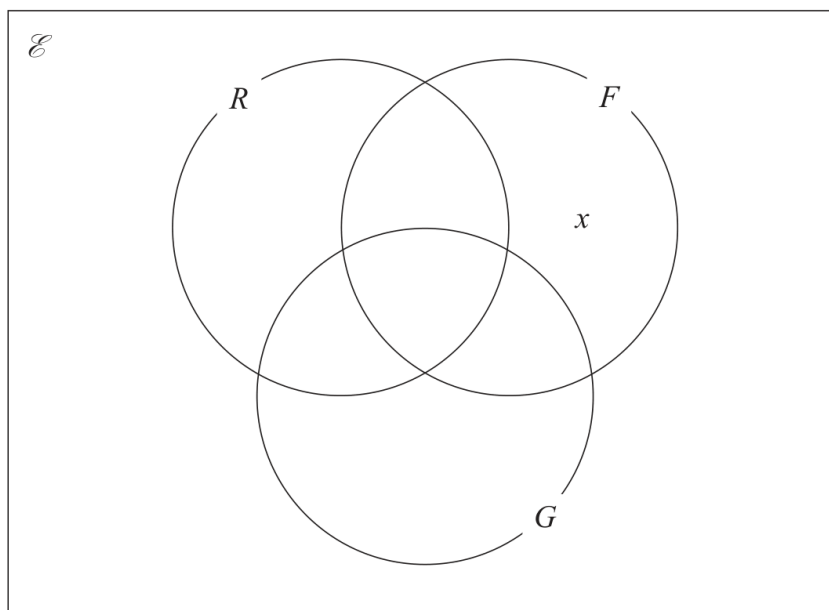
“Which of the subjects Russian (R), French (F) and German (G) do you study?”

Of these students

- 4 study all three of Russian, French and German
- 10 study Russian and French
- 13 study French and German
- 6 study Russian and German
- 24 study German
- 11 study none of the three subjects
- the number who study Russian only is twice the number who study French only.

Let x be the number of students who study French only.

(a) Show all this information on the Venn diagram, giving the number of students in each appropriate subset, in terms of x where necessary.



(3)

Given that the number of students who were asked the question was 80

(b) work out the number of these students that study Russian.

(3)

YOUR WORKING

16 Some students were asked the following question.

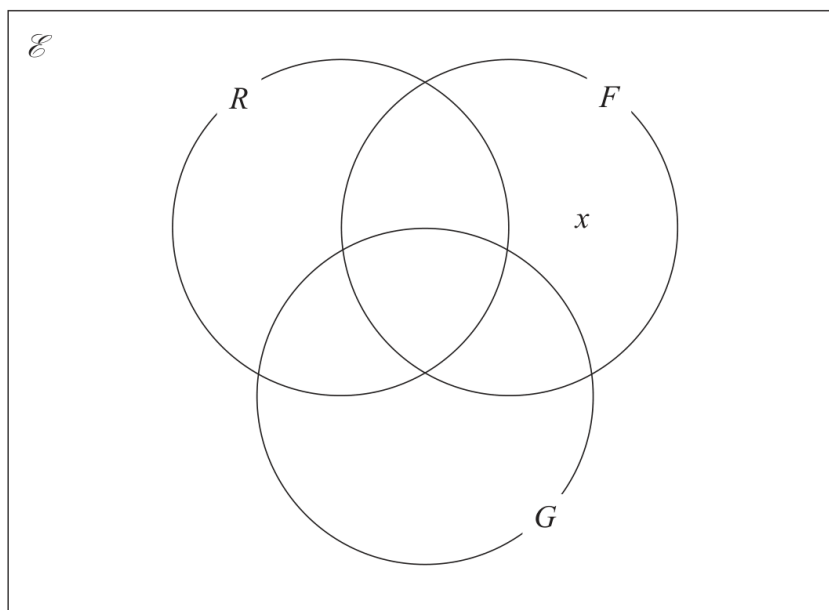
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(b) work out the number of these students that study Russian.

(3)

YOUR WORKING

(iii) $n([A \cap B] \cup C)$

(1)

YOUR WORKING

4MA1-2H Q18

2 marks

18 Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

(Total for Question 18 is 2 marks)

WORKING SPACE

- 18** $0.4\dot{x}$ is a recurring decimal.
 x is a whole number such that $1 \leq x \leq 9$

Find, in terms of x , the recurring decimal $0.4\dot{x}$ as a fraction.
Give your fraction in its simplest form.
Show clear algebraic working.

YOUR WORKING

16 Use algebra to show that $0.4\dot{3}\dot{8} = \frac{217}{495}$

(Total for Question 16 is 2 marks)

YOUR WORKING

4MA1-2H Q22

4 marks

22 The diagram shows a square inside rectangle $ABCD$

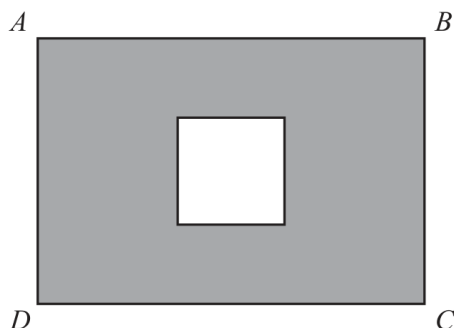


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X \text{ cm}^2$

$AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2 \text{ cm}$ correct to 2 significant figures

side of square = 4.1 cm correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

WORKING SPACE

4MA1-1H Q21 M01

3 marks

$$21 \quad T = \frac{x^2 + y^2}{w}$$

 $x = 28.4$ correct to 1 decimal place. $y = 17$ correct to 2 significant figures. $w = 90$ correct to the nearest 5Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.

Show your working clearly.

.....

WORKING SPACE

- 17** A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3}) \text{ m}^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

..... kg/m^3

(Total for Question 17 is 4 marks)

YOUR WORKING

19 The acceleration, a , of an object is given by

$$a = \frac{v - u}{t}$$

where

$v = 45.23$ correct to 2 decimal places

$u = 5.12$ correct to 2 decimal places

$t = 8.5$ correct to 2 significant figures

By considering bounds, work out the value of a to a suitable degree of accuracy.
Show your working clearly and give a reason for your answer.

$a = \dots\dots\dots$

(Total for Question 19 is 5 marks)

YOUR WORKING

4MA1-2H Q23

4 marks

- 23 The diagram shows a cuboid with a square cross section.

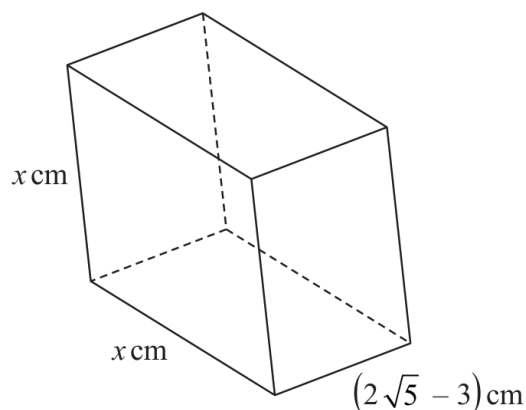


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x = \dots\dots\dots$

Continue your method**4 marks**

WORKING SPACE

4MA1-1H Q17 M01

1 marks

17 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

.....
(1)

WORKING SPACE

4MA1-1H Q16 Q16

3 marks

- 16 Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$
Show your working clearly.

WORKING SPACE

17 (a) Show that $(6 + 2\sqrt{12})^2 = 12(7 + 4\sqrt{3})$

Show each stage of your working.

(3)

YOUR WORKING

19 Without using a calculator, rationalise the denominator of $\frac{6}{3 - \sqrt{7}}$

Simplify your answer.

You must show each stage of your working.

(Total for Question 19 is 3 marks)

YOUR WORKING

4MA1-2H Q23

3 marks

23 Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

WORKING SPACE

4MA1-2H Q24 Q24

3 marks

24

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

WORKING SPACE

26 Find the values of n such that

$$\frac{10^{4n} \times 2^{3(n^2-5n)} \times 5^{2(1-2n)}}{20^2} = 1$$

Show clear algebraic working.

(Total for Question 26 is 5 marks)

YOUR WORKING

4MA1-2H Q16 whole

3 marks

16 Expand and simplify $(2x + 3)(x - 5)(x + 4)$

.....

WORKING SPACE

17 (a) Factorise $6y^2 - y - 5$

(2)

YOUR WORKING

4MA1-2H Q19

3 marks

19 Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

WORKING SPACE

4MA1-1H Q20 Q20(a), Q20(b)

4 marks

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

.....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$
The point *A* is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of *A*

(.....,)
(1)

WORKING SPACE

4MA1-2H Q24 whole

5 marks

24 Solve $\frac{45x^2 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$

WORKING SPACE

4MA1-2H Q23 M01

4 marks

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.

Show clear algebraic working.

WORKING SPACE

21 Write $\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

YOUR WORKING

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

YOUR WORKING

4MA1-2H Q22 Q22

7 marks

- 22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,
find the coordinates of the points B and D .

(..... ,)

(..... ,)

WORKING SPACE

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$

J is the point with coordinates $(j, 15)$ where $j < 0$

K is the point with coordinates $(6, k)$

M is the midpoint of JK .

The gradient of HM is 2

Find the value of j and the value of k .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

YOUR WORKING

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

YOUR WORKING

23 $ABCD$ is a kite.

$$AB = AD \text{ and } CB = CD$$

The point B has coordinates $(k, 1)$ where k is a negative constant.

The point D has coordinates $(8, 7)$

The straight line L passes through the points B and D

The straight line L is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC

Give your answer in the form $px + qy = r$ where p , q and r are integers.

Show your working clearly.



YOUR WORKING

(Total for Question 23 is 6 marks)

YOUR WORKING

4MA1-2H Q18

2 marks

18 The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

WORKING SPACE

19 (a) Write down an equation of a line that is parallel to the line with equation $y = 7 - 4x$

(1)

YOUR WORKING

- 20** The straight line **L** passes through point $A(-6, 2)$ and point $B(5, 3)$
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B .
The line **M** intersects the line $x = -1$ at point C .

Calculate the area of triangle ABC .

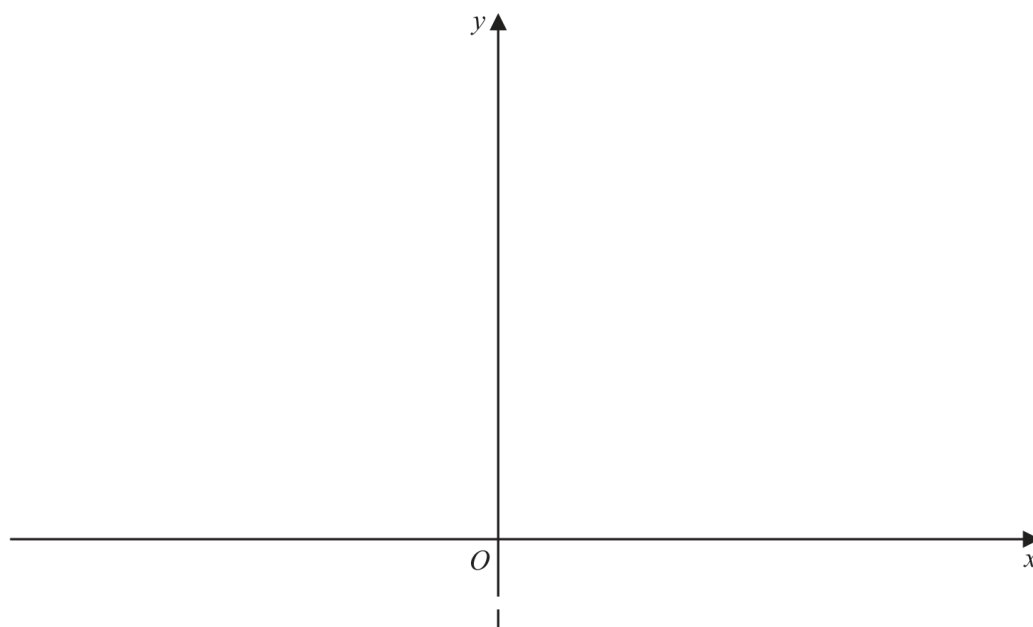
YOUR WORKING

4MA1-2H Q25

3 marks

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis.



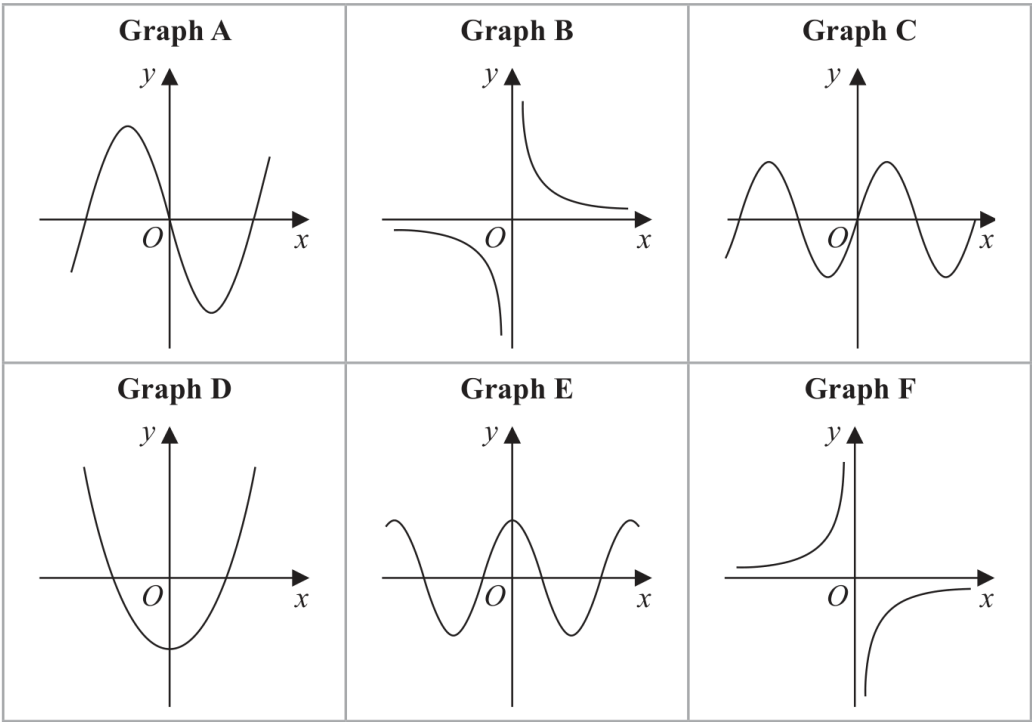
(3)

WORKING SPACE

4MA1-2H Q18

3 marks

18 Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \sin x$	
$y = -\frac{3}{x}$	
$y = 4x^3 - 5x$	

Continue your method**3 marks**

WORKING SPACE

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

YOUR WORKING

20

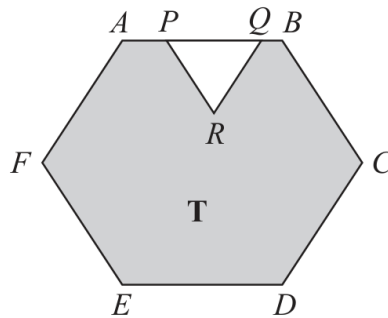


Diagram **NOT**
accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3} \text{ cm}^2$

work out the length of PQ .

..... cm

YOUR WORKING

4MA1-2H Q20 whole

4 marks

- 20 The diagram shows equilateral triangle ABC with sides of length 10 cm.
A circle is drawn inside the triangle.

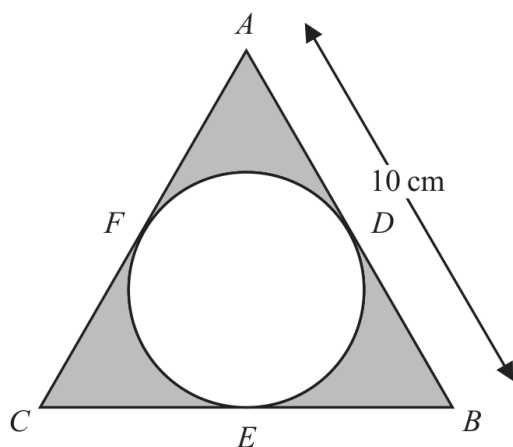


Diagram **NOT**
accurately drawn

D , E and F are points on the circle.

ADB , BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.

Give your answer correct to 3 significant figures.

..... cm^2

WORKING SPACE

Continue your method**4 marks**

WORKING SPACE

4MA1-1H Q22

6 marks

22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

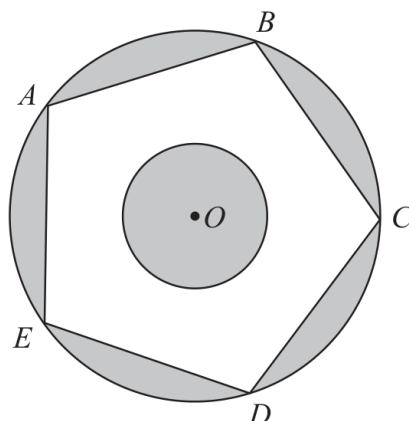


Diagram **NOT**
accurately drawn

A , B , C , D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

Continue your method

6 marks

WORKING SPACE

4MA1-1H Q16 M01

4 marks

16 $OAPB$ is a sector of a circle, centre O

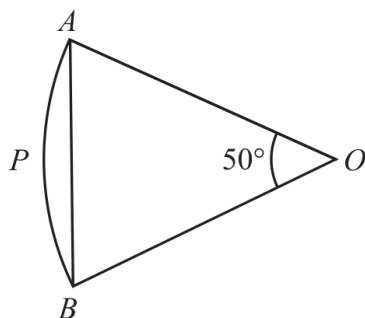


Diagram **NOT**
accurately drawn

Angle $AOB = 50^\circ$

Area of triangle $OAB = 120 \text{ cm}^2$

Work out the area of the sector $OAPB$

Give your answer correct to 3 significant figures.

..... cm^2

WORKING SPACE

4MA1-2H Q25 M01

5 marks

25 The diagram shows an equilateral triangle ABC and a circle with centre O

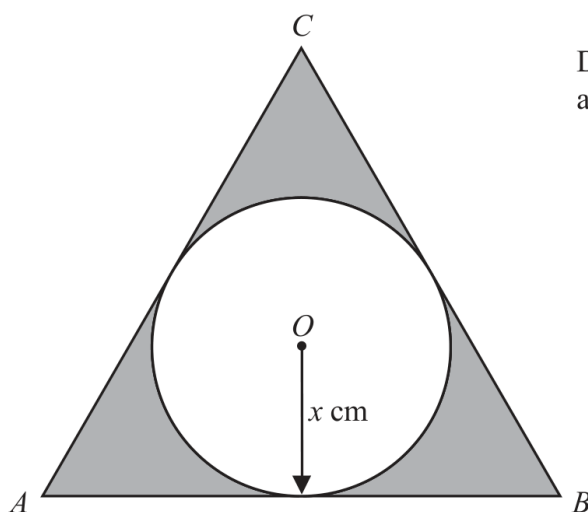


Diagram **NOT**
accurately drawn

AB , BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2 , of the regions shown shaded in the diagram is nx^2

Find the value of n

Give your answer correct to 3 significant figures.

4MA1-2H Q25 M01

5 marks

 $n = \dots\dots\dots$

WORKING SPACE

4MA1-1H Q16

4 marks

16 The diagram shows a circle with centre O

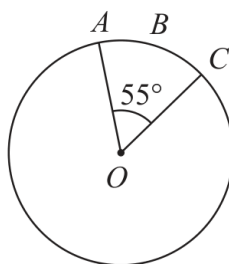


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm^2

Continue your method**4 marks**

WORKING SPACE

21 The diagram shows the cross section of a circular water pipe.

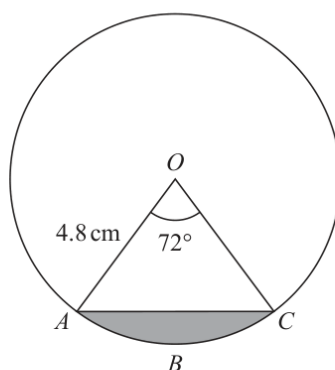


Diagram **NOT**
accurately drawn

OAC is a sector of the circle, centre O

The shaded region in the diagram represents the water flowing in the pipe.

The water flows at 14 cm/s in the pipe.

Work out the volume of water that has flowed through the pipe in 3 minutes.
Give your answer in cm^3 correct to 3 significant figures.



YOUR WORKING

..... cm³

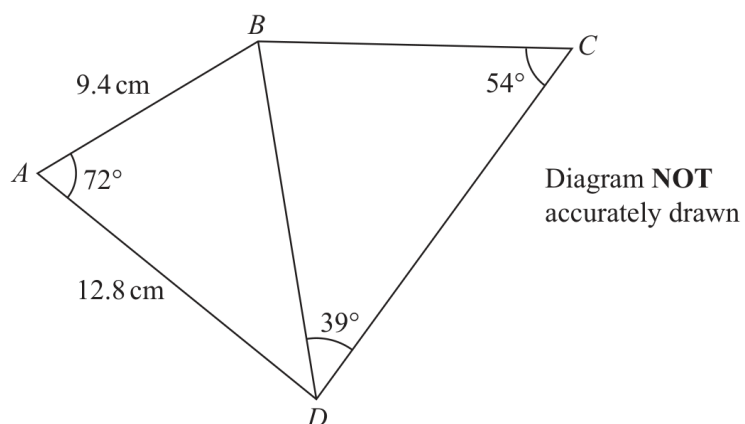
(Total for Question 21 is 5 marks)

YOUR WORKING

4MA1-1H Q20

5 marks

20



Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

WORKING SPACE

4MA1-2H Q22

5 marks

22 The diagram shows a triangle ABC and a flagpole BF

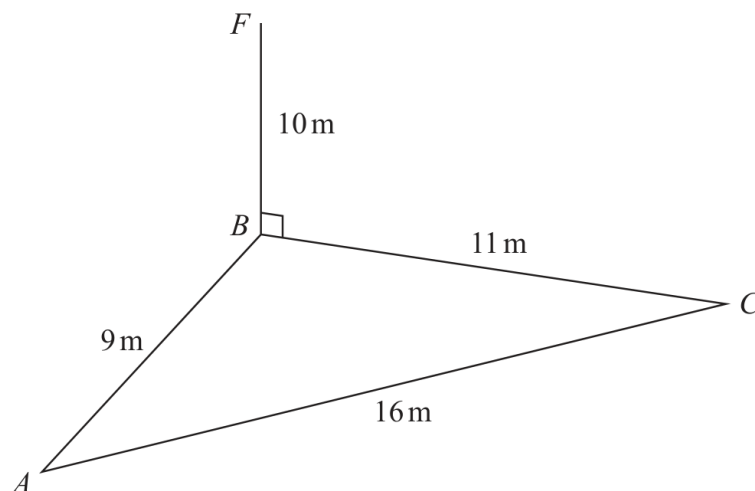


Diagram **NOT**
accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9 \text{ m} \quad BC = 11 \text{ m} \quad AC = 16 \text{ m} \quad BF = 10 \text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D
Give your answer correct to one decimal place.

○

Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q18 Q18

5 marks

18 Here is triangle ABC

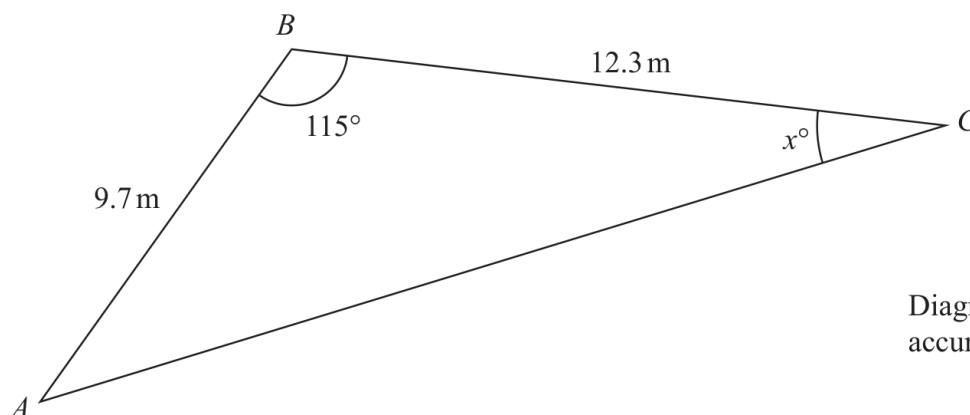


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x =$

Continue your method**5 marks**

WORKING SPACE

4MA1-2H Q23

6 marks

23 The diagram shows triangle PQR

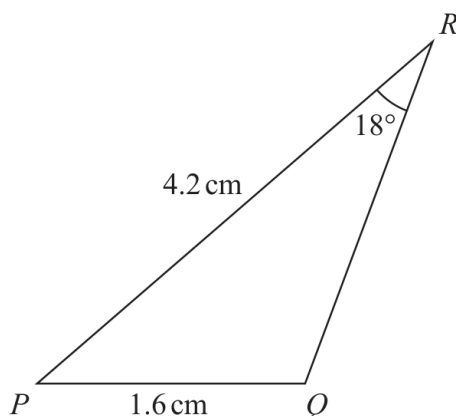


Diagram **NOT**
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

Continue your method

6 marks

WORKING SPACE

- 16 Here is a shape formed from two triangles ABC and CDE
 ACD and BCE are straight lines.

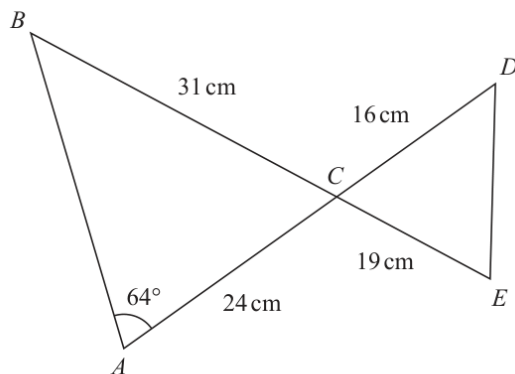


Diagram **NOT**
accurately drawn

$$AC = 24 \text{ cm} \quad BC = 31 \text{ cm} \quad CE = 19 \text{ cm} \quad CD = 16 \text{ cm}$$

$$\text{Angle } BAC = 64^\circ$$

Work out the length of DE

Give your answer correct to 3 significant figures.



YOUR WORKING

January 2023 | 1H | Question 16 | 5 marks

Chapter OL-T49: Sine, Cosine Rule & Area of Triangles

.....cm

(Total for Question 16 is 5 marks)

YOUR WORKING

4MA1-1H Q21

5 marks

- 21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

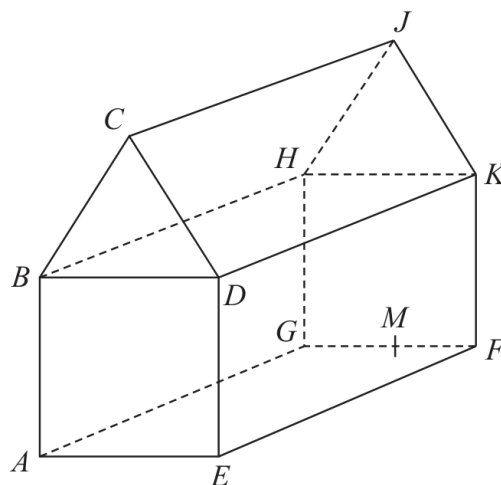


Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T = \dots\dots\dots$

Continue your method**5 marks**

WORKING SPACE

22 The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

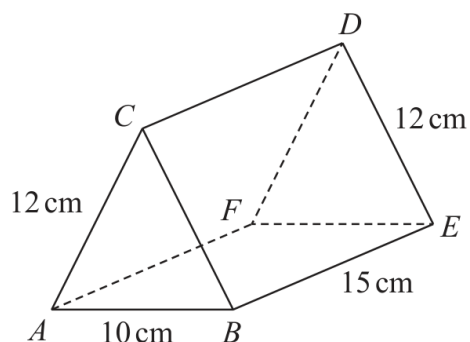


Diagram **NOT**
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between AD and the base $ABEF$.
Give your answer correct to 3 significant figures.

YOUR WORKING

18 The diagram shows cuboid $ABCDEFGH$.

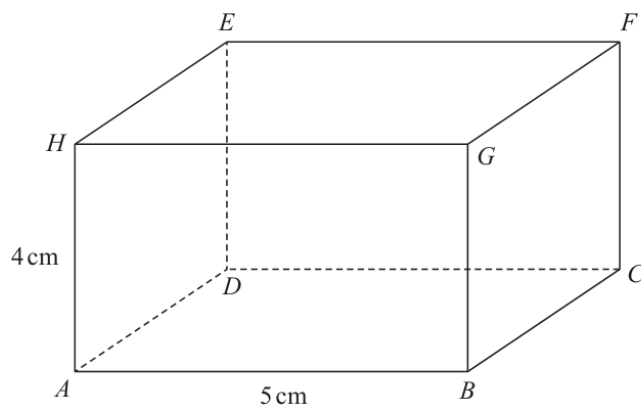


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm}$$

$$AH = 4 \text{ cm}$$

The size of the angle between CH and the plane $ABCD$ is 35°

Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

.....cm³

(Total for Question 18 is 5 marks)

YOUR WORKING

23 The diagram shows a solid prism $ABCDEFGHJ$

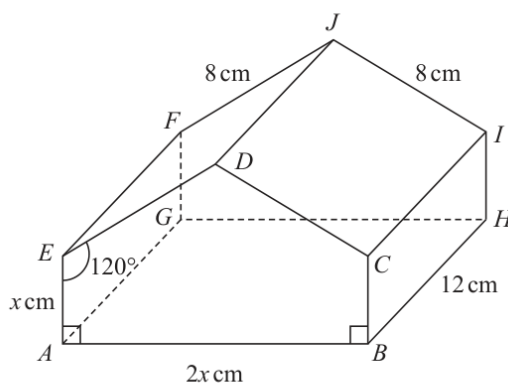


Diagram **NOT**
accurately drawn

The prism is such that each cross section is a pentagon where

$$AE = BC = x \text{ cm} \quad AB = 2x \text{ cm} \quad ED = CD = 8 \text{ cm}$$

$$\text{angle } EAB = \text{angle } CBA = 90^\circ \quad \text{angle } AED = \text{angle } BCD = 120^\circ$$

Given that $AG = BH = EF = DJ = CI = 12 \text{ cm}$

calculate the angle that AJ makes with the base $ABHG$ of the prism.

Give your answer correct to 3 significant figures.



YOUR WORKING

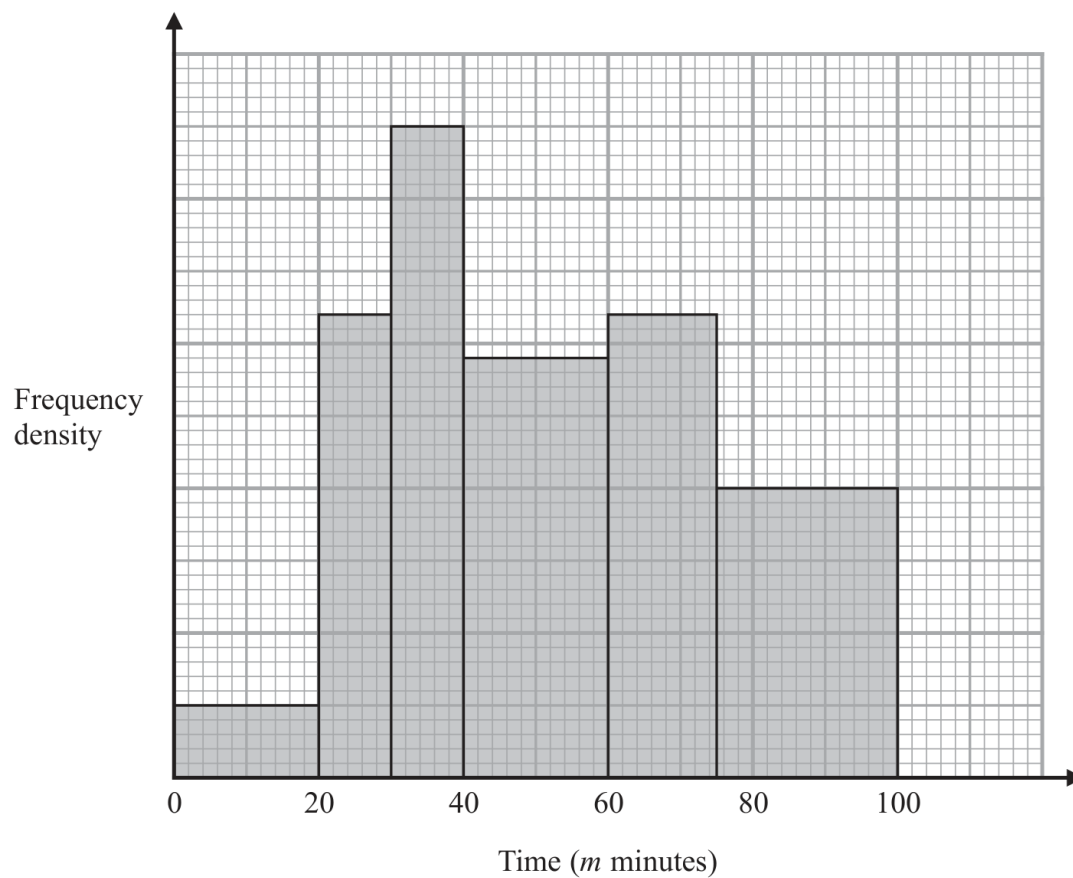
(Total for Question 23 is 5 marks)

YOUR WORKING

4MA1-1H Q21 Q21

3 marks

- 21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



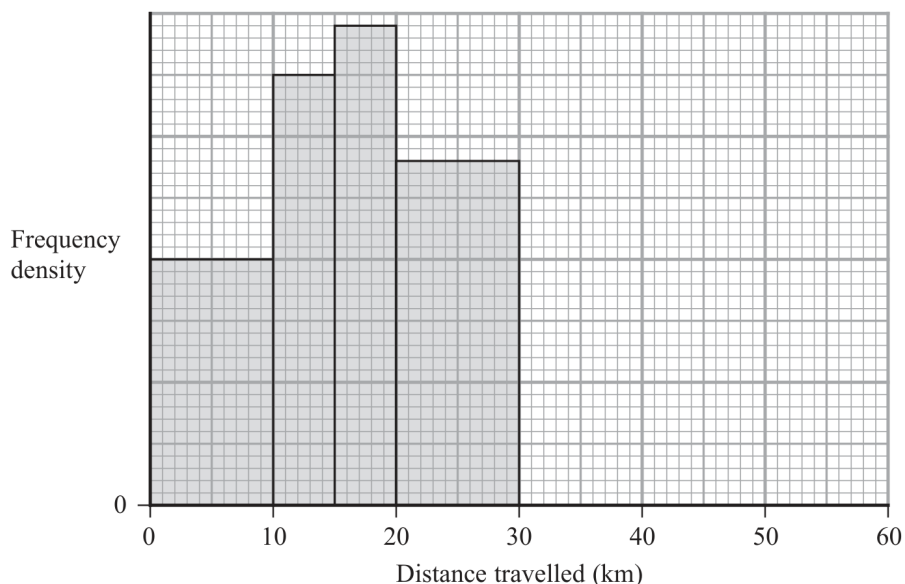
There are no children for whom $m > 100$

There are 10 children for whom $m \leq 20$

Work out an estimate for the number of children for whom $50 < m \leq 80$

- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
$20 \leq d < 30$	
$30 \leq d < 60$	30



Using the information in the table and in the histogram,

- (a) complete the table,

(2)

- (b) complete the histogram.

(1)

One of the people working in the store is chosen at random.

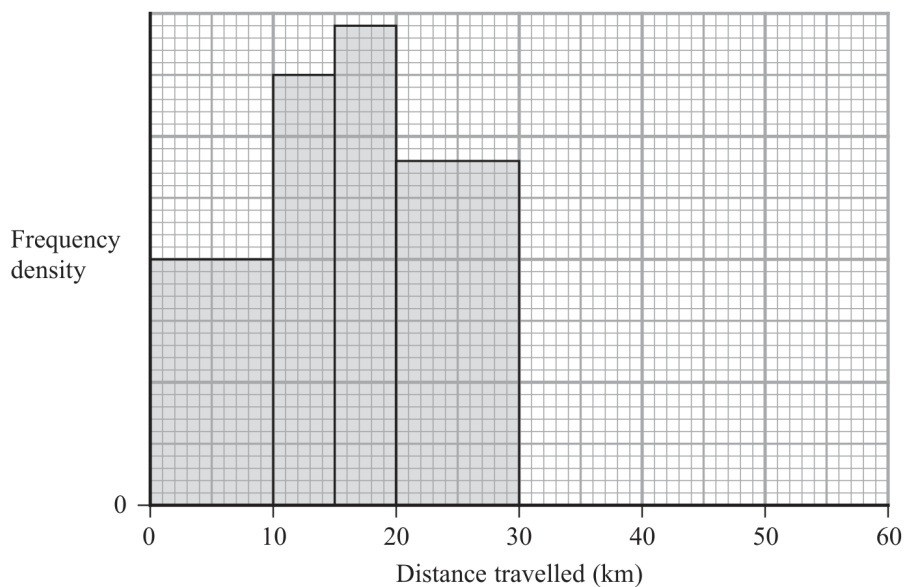
- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

(2)

YOUR WORKING

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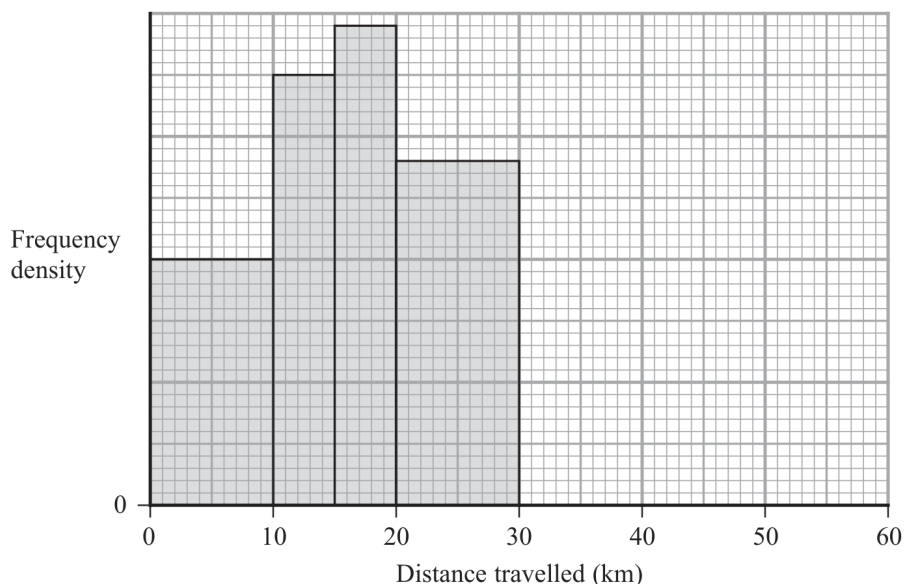
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YOUR WORKING

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(2)

YOUR WORKING

4MA1-2H Q16

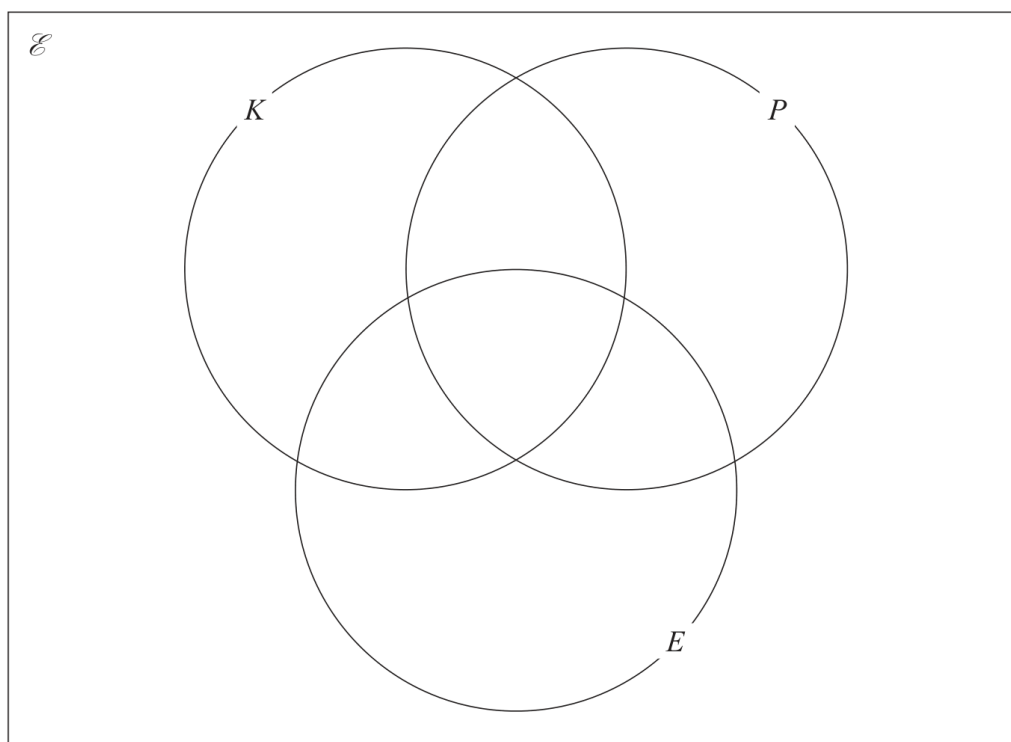
7 marks

- 16 60 art students were asked if they would like to attend workshops for knitting (K), for photography (P) or for embroidery (E)

Of these students

- 9 chose knitting, photography and embroidery
- 17 chose knitting and photography
- 16 chose photography and embroidery
- 20 chose knitting and embroidery
- 28 chose photography
- 39 chose embroidery
- 2 chose none of the workshops

- (a) Using this information, complete the Venn diagram to show the numbers of students in each subset.



(3)

4MA1-2H Q16

7 marks

One of the students is chosen at random.

Given that this student chose photography,

(b) find the probability that this student also chose knitting.

.....
(2)

(c) Find $n(P \cap K')$

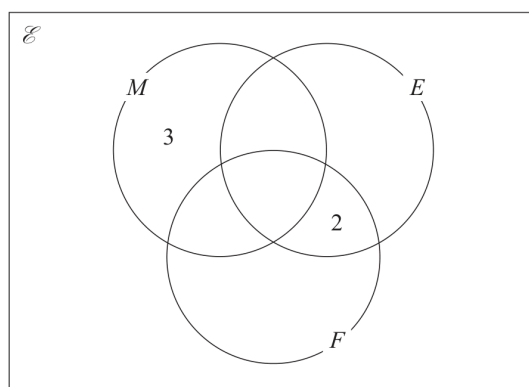
.....
(1)

(d) Find $n([P \cup E] \cap K)$

.....
(1)

- 16 There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
 25 students passed the test in Maths.
 20 students passed the test in English.
 14 students passed the test in French.
 18 students passed the tests in Maths and English.
 11 students passed the tests in Maths and French.
 4 students failed all three tests.
 x students passed all three tests.

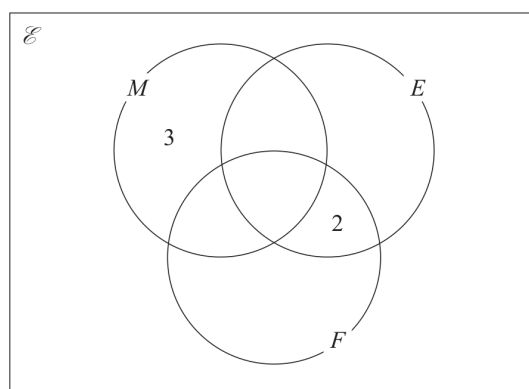
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$$x = \dots\dots\dots (2)$$

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

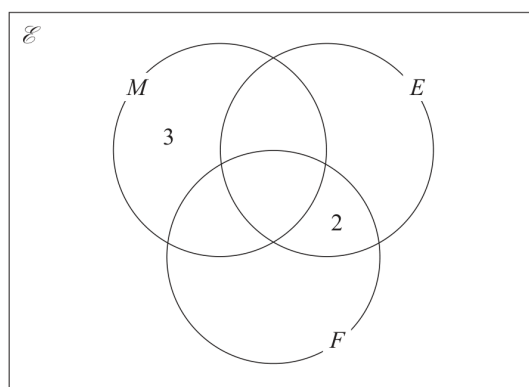
- (c) Find the probability that this student failed the test in French.

$$\dots\dots\dots (1)$$

YOUR WORKING

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 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
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 4 students failed all three tests.
 x students passed all three tests.

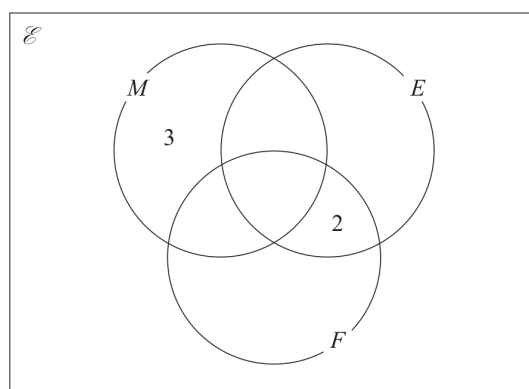
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- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$$x = \dots\dots\dots (2)$$

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

$$\dots\dots\dots (1)$$

YOUR WORKING

4MA1-1H Q21

3 marks

21 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

.....
(Total for Question 21 is 3 marks)

WORKING SPACE

4MA1-2H Q20

6 marks

20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots

12 grow carrots and potatoes and tomatoes

18 grow carrots and potatoes

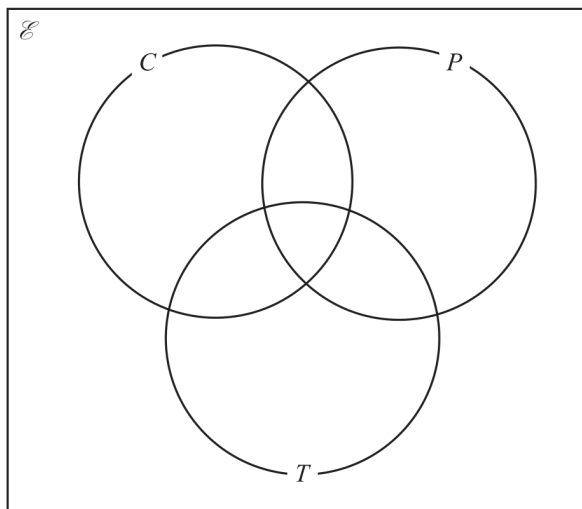
27 grow carrots and tomatoes

32 grow potatoes and tomatoes

29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

(2)

WORKING SPACE

4MA1-1H Q16

4 marks

16 There are 20 sweets in a box.

15 of the sweets are red

5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

WORKING SPACE

4MA1-1H Q20 demand

4 marks

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

.....

WORKING SPACE

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

8 marks

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers

31 have sheep

53 have chickens

6 have goats, sheep and chickens

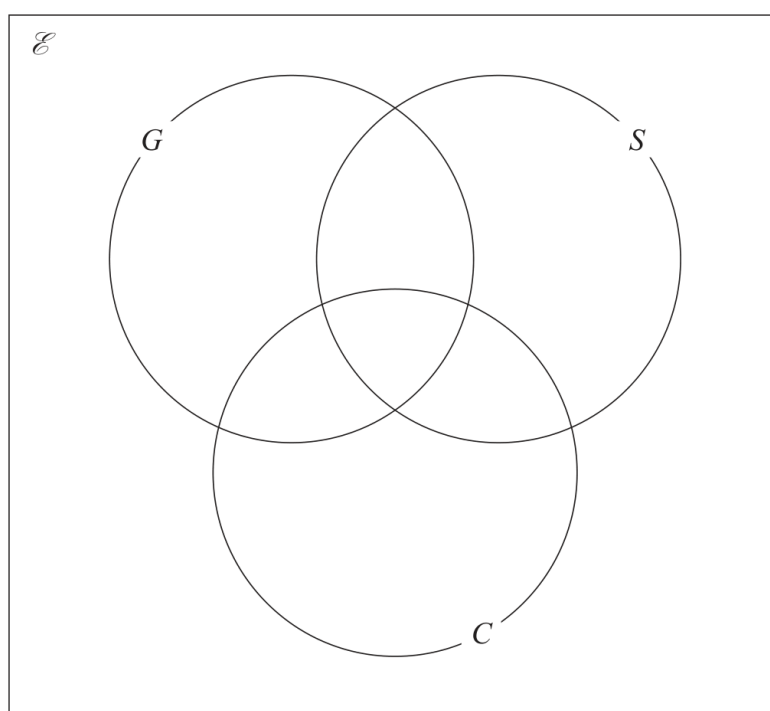
11 have sheep and goats

17 have sheep and chickens

18 have goats and chickens

20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



(3)

(b) Find

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

8 marks

(i) $n(G)$

(1)

(ii) $n([G \cup S]')$

(1)

(iii) $n(G' \cap C)$

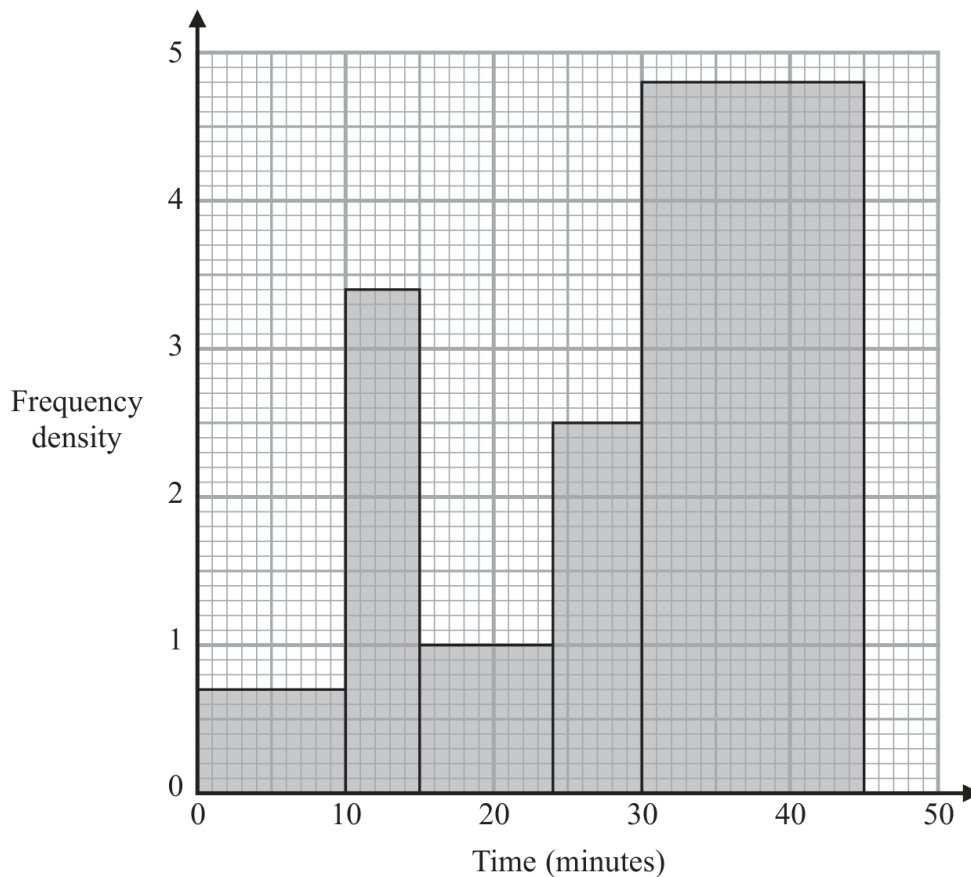
(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

(2)

- 18 The histogram gives information about the times, in minutes, that some customers spent in a supermarket.



One of the customers is selected at random.

Given that this customer had spent more than 30 minutes in the supermarket,

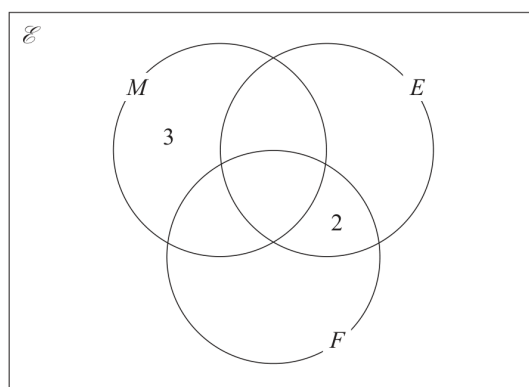
- (b) find the probability that this customer spent more than 36 minutes in the supermarket.

(2)

YOUR WORKING

- 16 There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
 25 students passed the test in Maths.
 20 students passed the test in English.
 14 students passed the test in French.
 18 students passed the tests in Maths and English.
 11 students passed the tests in Maths and French.
 4 students failed all three tests.
 x students passed all three tests.

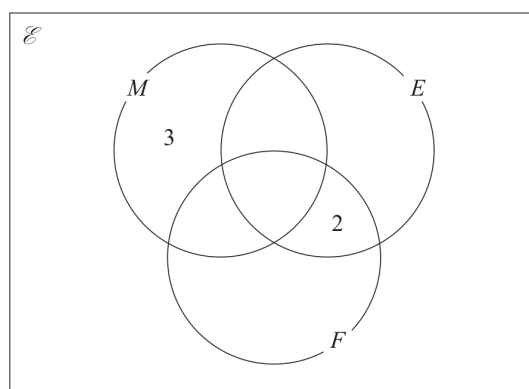
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$$x = \dots\dots\dots (2)$$

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

$$\dots\dots\dots (1)$$

YOUR WORKING

16 A box contains 15 counters.

There are 4 red counters, 5 green counters and the rest are yellow counters.

Niklas takes at random a counter from the box and writes down the colour of his counter.
He then puts the counter back into the box.

Sasha then takes at random a counter from the box and writes down the colour of her counter.

Work out the probability that the counters taken by Niklas and Sasha both have the same colour.

YOUR WORKING

20 A bag contains X counters.

There are only red counters and blue counters in the bag.

There are 4 more blue counters than red counters in the bag.

Finty takes at random 2 counters from the bag.

The probability that Finty takes 2 blue counters from the bag is $\frac{3}{8}$

Work out the value of X .

Show clear algebraic working.

(Total for Question 20 is 5 marks)

YOUR WORKING
